

```

{
    std::cout << "Hello Octave world!\n";
    int n = 2;
    Matrix a_matrix = Matrix (n, n);
    Matrix inv_a = Matrix(n,n);

    for (octave_idx_type i = 0; i < n; i++)
    {
        for (octave_idx_type j = 0; j < n; j++)
        {
            a_matrix(i, j) = (i + 1) * 10 + (j + 1);
        }
    }
    std::cout << a_matrix;

    inv_a = a_matrix.inverse();
    std::cout << "The inverse of the Matrix is: \n" << inv_a;

    return 0;
}

```

For all the API functions available, take a look at the Octave C++ API (the link is given in Resources, below).

Practicalities

Experienced programmers will realise that interfacing between two programming environments is rarely this simple in real-life

situations. Issues of memory allocation/deallocation and data interoperability usually creep in. Addressing these issues is beyond the scope of this article, which is about how to interface between C++ and Octave. C language programmers should consider using the (although deprecated) Mex-file interface. This is the same interface that MATLAB uses for interfacing with C code.

In next month's concluding part of this series, we will explore Octave packages and MATLAB compatibility. See you then. **END** 

Resources

- GNU Octave: <http://www.gnu.org/software/octave/>
- GNU Octave Documentation: <http://www.gnu.org/software/octave/docs.html>
- Good place to look for Oct-files: <http://octave.sourceforge.net/codas/>
- Octave C++ API: <http://octave.sourceforge.net/doxygen/html/>

Acknowledgements

HEVEA (<http://hevea.inria.fr/index.html>) was used for LATEX to HTML conversion.

By: Amit Saha

The author blogs at <http://amitsaha.wordpress.com> and welcomes queries and suggestions at amitsaha.in@gmail.com.

Summer Special

30% off for Students



RHCE | RHCSS



CCNA



SOLARIS



DBA

- **Pioneers in LINUX Training**
- **Certified Faculties**
- **Corporate Clients**
DELL, NRS, GENPACT ETC.,
- **Instructor Led Training**
- **1:1 Systems/ Routers**
- **Weekend Batches**
- **Placement Assistance**

New batches
03, 10, 17, 24 May 2010



Amrita Technologies
(Software Division-Hyderabad)

844/1, Shantiniketan Colony, Mahindra Hills,
East Marredpally, Secunderabad-26.

Tel: 9393733174 | 27733174 | 27737969

Email: aict.hybd@amrita.ac.in | www.amritahyd.org



SysAdmin Training by TUXG

**Want to become a better
Unix/Linux SysAdmin?**
Come join our training programs
and learn things the right way.

For practical IT exposure,
join our SAMIR program, a hands-on
learning experience that moulds you
for the IT industry.



TUXG Training
Training Wing of The UX Group

40-1-32, 1st Floor, Khanna Nagar
Benz Circle, VIJAYAWADA - 10
Call +91-866-6652358, +91-9989777151

<http://tuxgtraining.com>